

Detail-oriented and goal-driven biotechnology professional with a strong foundation in environmental science, sustainability research, and bioinformatics. Experienced in bioenzyme production from organic waste, waste-to-energy processes, and NGS data analysis using platforms such as Galaxy. Proficient in scientific reporting, life cycle and environmental impact assessments (LCA & EIA), and visualizing complex biological data through tools like Mermaid.js. Demonstrates excellent analytical skills, research acumen, and the ability to translate data into actionable insights. Backed by a strong academic record and hands-on internships, with a commitment to driving innovation in sustainable technologies and environmental conservation. Seeking to contribute technical expertise and research capabilities to forward-thinking organizations focused on green innovation and global sustainability

Education

Srm University Chennai

Btech - Biotechnology , CGPA: 8.8
KATTANKULATHUR , CHENNAI,

Aug 2022

Sree Ayyappa Public School

12th - PCB , CGPA: 9.2
BOKARO, JHARKHAND

Jul 2019

Ongoing projects and Events

Contributing Author – Book Chapter Development

Remote | May 2, 2025 – June 21, 2025

Working on a book chapter titled **Bio-based Monomers from Lignin: Synthesis and Applications.**

Start-Up Grand Challenge in Food Sector 3.0 – NIFTEM (DPIIT, MOFPI)

Jul 2025 – Present .

Experience

A Trainee – Sustainability & Waste Management

Steel Authority of India Limited (SAIL)

June 16, 2025

Bokaro, Jharkhand

Environmental Sustainability Intern – SAIL, Bokaro, Jharkhand
Interning in the Environment Department, focusing on industrial waste management and sustainability practices.
Gaining hands-on experience in on-site treatment processes, environmental compliance, and CPCB-based monitoring.

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Environmental Sustainability Intern

JAN 18 ,2025

CHENNAI, Tamil Nadu

SRM KTR, Chennai | Jan 2025

At the Sustainability Bootcamp, I actively contributed to showcasing a bioenzyme derived from flower waste as an eco-friendly solution for waste management and cleaning applications. I was involved in formulation optimization, data analysis, and impact assessment to ensure the product met sustainability standards. I also presented research findings, highlighting the biodegradability, efficiency, and environmental benefits of the bioenzyme. My

Skills

- Biotechnology
- Laboratory Techniques
- Research & Scientific Communication
- Sustainability & Environmental Analysis
- Technical & Computational Skills
- Problem-Solving & Innovation
- Teamwork & Time Management

Language

English – Full Professional Proficiency
Hindi – Native/Bilingual Proficiency

Technical Skills, Tools & Technologies
Programming & Scripting: HTML (Basic), JavaScript (Basic), Mermaid.js (Data Visualization)
Bioinformatics Tools: Galaxy (NGS Analysis), NCBI, BLAST
Data Analysis: Excel (Formulas, Pivot Tables, Charts), Basic R
AI Tools: Proficient in using AI-powered platforms for research assistance, content generation, data summarization, and visualization (e.g., ChatGPT, Gemini)
Technical Documentation: Scientific reporting, structured

Responsibilities included preparing technical reports and visual content to effectively communicate the concept, emphasizing circular economy principles and sustainable innovation.

D. Y. PATIL UNIVERSITY(Research Project)

Bioinformatics Research Intern

May 2024 - Jun 2024

AKRUDI ,PUNE

Akurdi, Pune | May 2024 – June 2024

As a Bioinformatics Research Intern, I worked on Next-Generation Sequencing (NGS) data analysis, focusing on preprocessing, sequence alignment, and variant analysis of large-scale genetic and microbial datasets. I utilized bioinformatics tools to interpret genetic variations and explore their environmental implications. My role involved analyzing and visualizing complex data, contributing to technical documentation, and presenting structured reports. This experience enhanced my skills in computational biology, data interpretation, and genomic research workflows.

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Industrial Training Intern – Waste Treatment & Vermicomposting

CHENNAI

Sep2024 - Sep 2024

- Gained hands-on experience in sewage treatment plant management and vermicomposting. Analyzed primary, secondary, and tertiary wastewater treatment processes, focusing on contaminant removal, sludge management, and microbial purification.
- Explored vermicomposting as a sustainable waste solution, studying organic decomposition, nutrient cycles, and soil enrichment. Strengthened expertise in wastewater treatment, resource recovery, and eco-friendly waste disposal strategies.

Awards

- PARTICIPANT CERIFICATE ON SUSTAINABILITY BOOTCAMP16/02/2025

Certifications

- Cancer Biology (Learn to upgrade)
- Forensic Science (Learn to upgrade)
- AI Dashboards using Microsoft Power Bi (Skill Nation)
- AI Beginners (HP LIFE)
- Bootcamp of sustainability (SRM ktr)
- ESG with sustainability technologies (Reliance foundation skill India)

Projects

Cleansing Bioenzymes from Flower Waste- SRMIST Chennai , Jan 2025 – Jun 2025 Produced

bioenzymes via fermentation of rose and marigold petals with jaggery and water. Utilized microbial enzymes (oxidoreductases, hydrolases, esterases) for natural stain removal. Formulated an eco-friendly, biodegradable soap enriched with bioenzymes and plant-based surfactants as a sustainable alternative to chemical detergents. Conducted constituent profiling and microbiome analysis to assess enzyme efficacy.

Next Generation Sequencing Data Analysis – Galaxy Platform

May 2024 – Jun 2024 Hands-on experience in NGS data analysis using Galaxy platform. Performed quality control, pipeline execution, and interpretation of raw genomic data.

Gained insights into genomic applications in personalized medicine, evolutionary biology, and medical research. Strengthened bioinformatics proficiency and analytical skills through workflow-based genomic analysis.

research presentation
Sustainability Tools: Life Cycle
Assessment (LCA),
Environmental Impact
Assessment (EIA)

Assessed greenhouse gas (GHG) emissions across major steel production units using IPCC guidelines and plant-specific operational data. Calculated emission baselines and identified high-emission hotspots. Benchmarked emission levels against global and national steel industry standards. Proposed targeted mitigation strategies, including energy optimization, waste heat recovery, and process improvements. Contributed to environmental reporting and compliance documentation as part of the Environment Department.